

Marshal Ashif Shawkat

+8801521119050 | marshalashif21@gmail.com | [Linkedin](#) | [GitHub](#)

RESEARCH INTERESTS

Medical Image Analysis, Explainable AI, Multimodal AI, Computer Vision

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Biomedical Engineering

Dhaka, Bangladesh

Feb. 2020 – March 2025

- **GPA:** 3.83/4.00 (Rank 7th in class)
- **Relevant Coursework:** Complex Variable and Vector Calculus, Computer Programming, Embedded Systems and Interfacing, Linear Algebra, Probability & Statistics, Physiological Control Systems, Digital Signal Processing, Random Signals and Processes, Medical Imaging, Bioinformatics

EXPERIENCE

Research Engineer

mHealth Lab, Bangladesh University of Engineering and Technology (BUET)

May 2025 – Present

Dhaka, Bangladesh

- Curated and organized a large-scale dataset of 100,000 chest X-ray images collected from [IEDCR](#).
- Collaborated with clinical radiologists to develop expert-annotated labels for chest X-ray datasets.
- Drove the successful pilot implementation of an in-house developed AI system in a local hospital.

PROJECTS

Tuberculosis Detection from Chest X-ray Images

Undergraduate Thesis | Supervisor: Dr. Taufiq Hasan, Professor, BME, BUET

Nov. 2023 – March 2025

[Presentation](#) | [Thesis book](#)

- Trained different types of neural networks, such as convolutional neural networks (VGG, ResNet, DenseNet) and vision transformers (ViT, Masked Autoencoder), both from scratch and through fine-tuning using PyTorch on a tuberculosis dataset.
- Visualized model attention on chest X-ray images during TB detection using Grad-CAM.
- Devised an algorithm to draw bounding boxes from heatmaps and implemented it using OpenCV.
- Calculated the largest eigenvalues to analyze the nature of the loss landscape.
- Computed centered kernel alignment (CKA) to measure similarity between models trained on the same dataset.
- Improved the TB representation of a CNN model by using soft labels from the knowledge distillation method and analyzed the relationship between improved representation and generalization capability on unseen TB datasets.
- Explored how randomness (e.g., weight initialization and batch data shuffling) impacts out-of-distribution (OOD) performance.

GPT-2 Pretraining from Scratch

Personal project

Nov. 2025

[GitHub](#)

- Tokenized the Book Corpus dataset using Tiktoken and implemented the GPT-2 model using PyTorch.
- Utilized 4× NVIDIA GPUs on a cloud platform with PyTorch DistributedDataParallel (DDP) for scalable multi-GPU training.
- Developed a custom DataLoader for efficient dataset handling.
- Applied modern techniques like mixed precision training, torch compile, fused AdamW, flash attention, gradient accumulaton etc. to speedup the training process.

Brain Tumor Classification from Magnetic Resonance Images

PI: Shoyad Ibn Sabur Khan Nuhash, Assistant Professor, BME, BUET

Feb. 2023

[GitHub](#)

- Implemented a custom model [BrainMRNet](#) from scratch.
- Gained hands-on experience with key deep learning concepts, including attention modules, residual blocks, and hypercolumn techniques to improve model performance.
- Compared the performance of the custom BrainMRNet model against other established architectures (VGG16, ResNet50, DenseNet121) using metrics such as accuracy, AUC, sensitivity, and specificity.
- Developed an interactive Streamlit-based web application enabling local inference: users upload MRI images, the model processes them, and results are displayed in real-time.

ECG Signal Filtering and Heart Rate Measurement

Sep. 2023

PI: Dr. Md. Kamrul Hasan, Professor, EEE, BUET

Presentation

- Processed raw ECG signals from the MIT-BIH Arrhythmia Dataset.
- Explored filtering techniques (Butterworth, Chebyshev Type II, IIR Notch, and moving average) to remove baseline wander, powerline interference, and EMG noise from raw ECG signals.
- Applied wavelet transform for accurate QRS complex detection and heart rate estimation.
- Developed a MATLAB GUI application for ECG signal denoising and heart rate measurement.

Block-Based Hair Removal in Dermoscopic Images

Mar. 2024

PI: Samiul Based Shuvo, Assistant Professor, BME, BUET

Presentation

- Implemented and evaluated a block-based algorithm for automatic hair removal in dermoscopic images to improve skin lesion classification.
- Converted RGB images into YIQ color space and applied morphological bottom-hat filtering for hair detection.
- Performed block-wise inpainting using histogram-based pixel replacement followed by morphological closing to reconstruct lesion regions.
- Achieved comparable performance to the reference method (Dull Razor) while maintaining effectiveness across diverse skin tones and varying hair densities.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++, R, JavaScript (*in order of proficiency*)

ML Frameworks: PyTorch, TensorFlow/Keras, PyTorch Geometric, JAX (*in order of proficiency*)

Python Libraries: NumPy, pandas, Matplotlib, scikit-learn, OpenCV, Pillow, SciPy, Beautiful Soup, Streamlit

Software & Tools: GNU/Linux, Bash scripting, LaTeX, Git version control

Embedded Systems & Hardware: Atmega32 Microcontroller, Arduino, Raspberry Pi

CAD & Simulations: SOLIDWORKS, FreeCAD, Ansys, COMSOL Multiphysics, Proteus, TINA-TI

PUBLICATIONS

Shawkat, M. A., Hasan, M., & Hasan, T. (2025). Weakly supervised tuberculosis localization in chest X-rays through knowledge distillation. [Preprint available in arXiv (<https://arxiv.org/abs/2512.11057>)]

LANGUAGE

Bengali (Native), English (IELTS Band 8.0)

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

Vice President | *BUET Biomedical Engineering Society (BMES) Student Chapter*

Sep. 2024 – May 2025

- Organized a nationwide scientific article writing competition engaging college and university students.
- Coordinated seminars and outreach programs on topics ranging from breast cancer awareness to student mental health.

Science Writer | *The Royal Scientific Publications Ltd*

Aug. 2022 – Present

- Founded a monthly scientific magazine Divention that reached over 10,000 high school & college students.
- Authored 40+ articles on diverse topics including artificial intelligence and neuroscience to foster scientific curiosity among young readers

Class Representative | *Bangladesh University of Engineering and Technology*

Nov. 2022 – Nov. 2023

- Proactively served as a liaison between students and faculty, effectively coordinating with professors to optimize class schedules and examination dates for enhanced academic efficiency.
- Advocated for student needs by addressing and resolving critical classroom infrastructure issues, including sound systems and air conditioning

AWARDS & ACHIEVEMENTS

- University Merit Scholarship, BUET
- Regional Winner, Bangladesh Junior Science Olympiad
- 1st prize, Regional Math Olympiad